

Requirements / Specification Document

TITLE: Type N and 2.4 mm Power Transfer Standards

Requesting Laboratory / Division / Group:

Communications Technology Laboratory
Radio Frequency Technology Division
Guided Wave Electromagnetics Group (672.01)

General Statement of Need:

The Communications Technology Laboratory (CTL) at the National Institute of Standards and Technology (NIST) enables science and industry by providing essential measurement methods, instrumentation, and standards to support all phases of communications technology development, from fundamental characterization and discovery to production.

To support research and measurement services in radio frequency and microwave power, the Radio Frequency Technology Division in CTL requires Type N and 2.4 mm power transfer standards that are compatible with microcalorimeters. These items are being acquired as part of an agreement with the Navy Primary Standards Laboratory.

Item Specifications and Quantities:

Item 1: Type N Power Transfer Standard, Impedance-Matched		
Quantity: 3 Each		
Specifications:	1.1	Shall have a 50 Ω Type N coaxial interface.
	1.2	Shall have a radio-frequency (RF) termination with an operating frequency range from DC to 18 GHz.
	1.3	The RF termination shall have a reflection coefficient magnitude of less than 0.10 across the operating frequency range when referenced to a 50 Ω impedance.
	1.4	The RF termination shall have a maximum input power of at least 20 dBm.
	1.5	Shall have a DC heater with a maximum input power of at least 20 dBm.
	1.6	The DC heater shall have a four-wire lead configuration.
	1.7	The DC heater shall have a DC resistance in the range of 400 Ω to 600 Ω .
	1.8	Shall have a thermopile to sense heating from power dissipation in the RF termination or the DC heater.

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1.9	The sensitivity of the thermopile shall be at least 150 mV/W at DC.
1.10	Shall have a negative temperature coefficient (NTC) resistor for temperature determination.
1.11	Shall have a DC connector that allows direct electrical connections to the DC heater, thermopile, and NTC resistor.
1.12	If the transfer standard contains active readout or data storage electronics, it shall be possible to operate the transfer standard without supplying power to these electronics. The RF termination, DC heater, and thermopile, and NTC resistor shall be operable without supplying power to active readout or data storage electronics.
1.13	Shall have dimensions of less than 90 mm × 30 mm × 30 mm (length × width × height).
1.14	Shall have a total mass of less than 150 g.
1.15	Shall include a 1.5 m cable that is mechanically and electrically compatible with the DC connector of the sensor.

Item 2:	Type N Power Transfer Standard, Open Circuit
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Quantity:	1 Each
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Specifications:	2.1	Shall have the same design as the Type N impedance-matched power transfer standard, with the exception that the element that dissipates RF power is replaced with an open-circuit termination designed to reflect RF power.
	2.2	Shall include a 1.5 m cable that is mechanically and electrically compatible with the DC connector of the sensor.

Item 3:	Type N Power Transfer Standard, Short Circuit
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Quantity:	1 Each
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Specifications:	3.1	Shall have the same design as the Type N impedance-matched power transfer standard, with the exception that the element that dissipates RF power is replaced with a short-circuit termination designed to reflect RF power.
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	3.2	Shall include a 1.5 m cable that is mechanically and electrically compatible with the DC connector of the sensor.
Item 4:	Type N Power Transfer Standard, Mismatch Circuit	
Quantity:	1 Each	
Specifications:	4.1	Shall have the same design as the Type N impedance-matched power transfer standard, with the exception that the element that dissipates RF power is replaced with a is replaced with a termination designed to partially reflect RF power and partially dissipate RF power.
	4.2	Shall include a 1.5 m cable that is mechanically and electrically compatible with the DC connector of the sensor.
Item 5:	2.4 mm Power Transfer Standard, Impedance-Matched	
Quantity:	3 Each	
Specifications:	5.1	Shall have a 50 Ω 2.4 mm coaxial interface.
	5.2	Shall have a radio-frequency (RF) termination with an operating frequency range from DC to 50 GHz.
	5.3	The RF termination shall have a reflection coefficient magnitude of less than 0.15 across the operating frequency range when referenced to a 50 Ω impedance.
	5.4	The RF termination shall have a maximum input power of at least 20 dBm.
	5.5	Shall have a DC heater with a maximum input power of at least 20 dBm.
	5.6	The DC heater shall have a four-wire lead configuration.
	5.7	The DC heater shall have a DC resistance in the range of 400 Ω to 600 Ω .
	5.8	Shall have a thermopile to sense heating from power dissipation in the RF termination or the DC heater.
	5.9	The sensitivity of the thermopile shall be at least 150 mV/W at DC.
	5.10	Shall have a negative temperature coefficient (NTC) resistor for temperature determination.

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5.11	Shall have a DC connector that allows direct electrical connections to the DC heater, thermopile, and NTC resistor.
5.12	If the transfer standard contains active readout or data storage electronics, it shall be possible to operate the transfer standard without supplying power to these electronics. The RF termination, DC heater, and thermopile, and NTC resistor shall be operable without supplying power to active readout or data storage electronics.
5.13	Shall have dimensions of less than 90 mm × 30 mm × 30 mm (length × width × height).
5.14	Shall have a total mass of less than 120 g.
5.15	Shall include a 1.5 m cable that is mechanically and electrically compatible with the DC connector of the sensor.

Item 6:	2.4 mm Power Transfer Standard, Open Circuit
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Quantity:	1 Each
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Specifications:	6.1	Shall have the same design as the 2.4 mm impedance-matched power transfer standard, with the exception that the element that dissipates RF power is replaced with an open-circuit termination designed to reflect RF power.
	6.2	Shall include a 1.5 m cable that is mechanically and electrically compatible with the DC connector of the sensor.

Item 7:	2.4 mm Power Transfer Standard, Short Circuit
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Quantity:	1 Each
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Specifications:	7.1	Shall have the same design as the 2.4 mm impedance-matched power transfer standard, with the exception that the element that dissipates RF power is replaced with a short-circuit termination designed to reflect RF power.
	7.2	Shall include a 1.5 m cable that is mechanically and electrically compatible with the DC connector of the sensor.

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Item 8:	2.4 mm Power Transfer Standard, Mismatch Circuit	
Quantity:	1 Each	
Specifications:	8.1	Shall have the same design as the 2.4 mm impedance-matched power transfer standard, with the exception that the element that dissipates RF power is replaced with a is replaced with a termination designed to partially reflect RF power and partially dissipate RF power.
	8.2	Shall include a 1.5 m cable that is mechanically and electrically compatible with the DC connector of the sensor.

Vendor Requirements

1. Possess at time of award/contracting an Export Control Classification Number (ECCN).
2. Possess at time of award/contracting a Unique Entity Identification Number (UEI) which registration is listed as "Active" on the GSA System for Award Management (SAM.gov).

Procurement Support Information and Ancillary Services:

[Provide input for each section Below or Mark Not Applicable (N/A)]

Section	Requirements/Specifications
Planning Considerations ☒ N/A	☐ List any visits required to allow potential vendors to see site specific constraints. [Explain Requirements.] ☐ US citizen required on-site (foreign nationals require pre-registration 30 days prior to visit.) ☐ Vendor Representative On-Site more than three (3) days. ☐ Other Considerations: [List other considerations to ensure an accurate quote.]
Shipping & Delivery ☐ N/A	Shipping Address: ☒ NIST-Boulder ATTN: Provided at award Mail Stop: 325 Broadway

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Boulder, CO 80305

- ☐ NIST-Gaithersburg
ATTN:
Mail Stop:
100 Bureau Dr
Gaithersburg MD 20899
- ☐ [Alternate Shipping Address]

Shipping Criteria:

☒ Partial Delivery Acceptable

- ☐ [Enter Shipping and Delivery information by item.]
- ☐ Direct Delivery to OU Building Required
[Explain Need for Requirement]
Building /Room Number: [Enter Building/Room Number]
☐ Building has Loading Dock
Containerization Preference: [Enter Container Type]
Other Requirements:
[List Any Other Requirement]

Delivery Date Criteria:

☒ Delivery Shall Be Completed No Later Than:
4 months after contract award date

- ☐ Delivery Shall Be Made No Earlier Than:
Click or tap to enter a date.
- ☐ Scheduled Deliveries:

Item Number	Due Date

- ☐ Other:
[List Other Shipping, Delivery or Special Requirements.]

Electronic Media

[Enter Electronic Media Requirements by Item.]

☒ N/A

- ☐ Software
- ☐ Electronic Manuals
- ☐ License Key
- ☐ Other:

[List Any Other Electronic Media Requirements]

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- ☐ US Government Email Address for Software Delivery:
[Enter Address]

Installation

[Enter Installation Requirements by Item]

☒ N/A

- ☐ Vendor Technician on Site
- ☐ Rigging
- ☐ Uncrating / Unpackaging
- ☐ Removal of Packaging Material
- ☐ Equipment Set Up
- ☐ Start-Up Services
- ☐ Turnkey Installation
- ☐ Other:

[Enter Details]

NOTE: TPOC shall be the coordination POC.

Facility / Utility Considerations

[List Site Specific Utility Considerations by Item]

☒ N/A

- ☐ OFPM Work Order Numbers
[List Work Order Numbers]
- ☐ Power Available
- Voltage: [Enter voltage] (V)
- Maximum Current: [Enter max current draw] (A)
- Phase: [Choose Phase]
- Frequency: [Typically 60Hz] (Hz)
- NEMA Plug Type: [Enter Plug Type] (i.e.: NEMA L5-20P)
- Environment: [Choose Environment]
- Other Power Considerations:
[List Any Other Power Considerations.]
- ☐ Utility Connections:
[List Relevant Available Utilities, Pressure(s) and Flowrate(s).]
- ☐ Other Considerations:
[List Other Considerations to Ensure an Accurate Quote.]

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Inspection and
Acceptance

☐ N/A

[Inspection Requirements Per Item (Who, When, What, Where)]

☐ On-Site

☒ Acceptance Not Expected to Exceed 7 Business Days

☐ Acceptance Expected to Exceed 7 Business Days:
[Duration]

☐ Inspection and Acceptance Plan:
[Explanation]

☐ Other:
[Enter Other Requirements]

Training

☒ N/A

Training Requirements (Who, When, What, Where)

☐ Number of Trainees: [Enter Number of Trainees]

☐ On-Site

☐ Off-Site Location: [Enter Location(s)]

Is Trainee Travel Required: [Choose Answer]

Scope of Training

☐ Operation ☐ Maintenance ☐ Troubleshooting

☐ Safety ☐ Other

[Enter Other Scope of Training Requirements]

☐ Training Materials to be Provided by Government:
[Describe Materials]

☐ Training Materials to be Provided by Vendor:
[Describe Materials]

☐ Other
[Enter Other Training Requirements]

NOTE: SME shall be coordination POC.

Warranty

☐ N/A

Base Manufacturer Warranty is acceptable

Warranty Length: [i.e. 3 - Years]

What is included?

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- ☐ Labor ☐ Parts ☐ Travel
- ☐ Cost and Liability for Returns
- ☐ Other
[Enter Other Requirements]
- ☐ 1 Option Period
- ☐ 2 Option Periods
- ☐ 3 Option Periods
- ☐ 4 Option Periods (Maximum)

Calibration

☒ N/A

[Describe Calibration Requirements Per Item]

Calibration Plan Length: [i.e. 3 - Years]

What is Included?

- ☐ Labor ☐ Spare Parts ☐ Consumables
- ☐ Travel ☐ Cost and Liability for Shipment
- ☐ Software Updates ☐ Manuals
- ☐ Other
[Enter Other Requirements]
- ☐ Requirements:
[Enter Requirements] (i.e.: On-Site Within 72 hours, etc.)
- ☐ 1 Option Period
Length: [i.e. 1 – Year / 6 - Months]
- ☐ 2 Option Periods
- ☐ 3 Option Periods
- ☐ 4 Option Periods (Maximum)

Maintenance

☒ N/A

[Describe Maintenance Requirements Per Item]

Maintenance Plan Length: [i.e. 3 - Years]

What is Included?

- ☐ Labor ☐ Spare Parts ☐ Consumables
- ☐ Travel ☐ Cost and Liability for Shipment
- ☐ Software Updates ☐ Manuals
- ☐ Other
[Enter Other Requirements]
- ☐ Requirements:

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[Enter Requirements] (i.e.: On-Site Within 72 hours, etc.)

☐ 1 Option Period

Length: [i.e. 1 – Year / 6 - Months]

☐ 2 Option Periods

☐ 3 Option Periods

☐ 4 Option Periods (Maximum)

Government
Furnished
Material /
Equipment

☒ N/A

List Any Government Furnished Material / Equipment Per Item]

☐ Samples for Testing

☐ Equipment which will leave Government Site

Description: [Enter Description]

Serial Number: [Enter Serial Number]

NIST Property Number: [Enter NIST Property Number]

☐ Other: [Enter Other Details]

Travel

☒ N/A

[Describe any Travel Requirements Per Item]

☐ Explain:

[Enter Justification / Requirement]

☐ Other:

[Enter Other Requirements]

NOTE TO VENDOR: To the maximum extent possible, include travel requirement estimates as part of pertinent sections above (Installation, Maintenance, Repair, etc.). Otherwise, Travel will be a separate line item, reimbursable per the Federal Travel Regulation.